

TrES-3b

R-2821

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_210806 · created 2026-09-05T02:32:34+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459432.8621 +/- 0.0016 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.139 +/- 0.055
Transit depth (Rp/Rs)^2	1.93 +/- 1.54 [%]
Orbital Inclination (inc)	77.23 +/- 2.9
Airmass coefficient 1 (a1)	1.55 +/- 0.43
Airmass coefficient 2 (a2)	-0.314 +/- 0.077
Scatter in the residuals of the lightcurve fit is	21.48 %
Best Comparison Star	#1 - [493, 337]
Optimal Aperture	2.86
Optimal Annulus	11.81
Transit Duration (day)	0.009 +/- 0.019

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.139 $\pm$ 0.055 vs 0.16309 (deviation 0.41 σ)
Duration fit vs published	0.009 d vs 0.0567 d (deviation 2.37 σ)
Mid-time O-C	0.1 min
Depth SNR	2.4
Residual scatter	21.48 %

Light curve

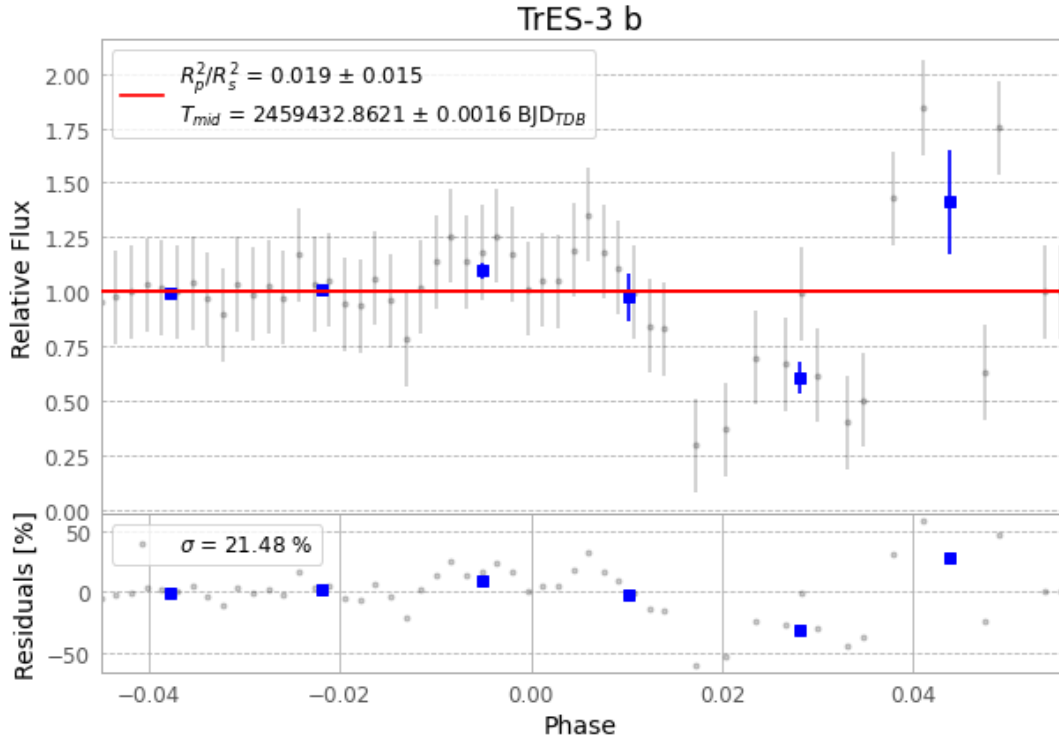


Figure 1 — FinalLightCurve_TrES-3 b_2021-08-06.png (SHA-256 c8c05b4c366a22c7...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [406, 248], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 dd19d9f29b95992c...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

64 FITS frames (42 MB), TRES-3210806071212.FITS → TRES-3210806102112.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-210806.log (959284b5a07a...), FinalLightCurve_TrES-3 b_2021-08-06.png (c8c05b4c366a...), FinalLightCurve_TrES-3 b_2021-08-06.csv (3da336f29c2c...)

Compute resources

Wall time 520.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-09-05 02:32Z.